A tomato-rich diet is related to depressive symptoms among an elderly population aged 70 years and over: a population-based, cross-sectional analysis.

BACKGROUND: Enhanced oxidative stress or defective anti-oxidant defenses are related to the pathogenesis of depressive symptoms. Lycopene is the most powerful antioxidant amongst the carotenoids. The aim of this study was to investigate the relationship between different vegetables, including tomatoes/tomato products (a major source of lycopene), and depressive symptoms in a community-based elderly population. METHODS: We analyzed a cross-sectional survey including 986 community-dwelling elderly Japanese individuals aged 70 years and older. Dietary intake was assessed using a valid self-administered diet-history questionnaire, and depressive symptoms were evaluated using the 30-item Geriatric Depression Scale with 2 cut-off points: 11 (mild and severe) and 14 (severe) or use of anti-depressive agents. RESULTS: The prevalence of mild and severe and severe depressive symptoms was 34.9% and 20.2%, respectively. After adjustments for potentially confounding factors, the odds ratios of having mild and severe depressive symptoms by increasing levels of tomatoes/tomato products were 1.00, 0.54, and 0.48 (p for trend <0.01). Similar relationships were also observed in the case of severe depressive symptoms. In contrast, no relationship was observed between intake of other kinds of vegetables and depressive symptoms. LIMITATIONS: This is a cross-sectional study, and not for making a clinical diagnosis of depressive episodes. CONCLUSIONS: This study demonstrated that a tomato-rich diet is independently related to lower prevalence of depressive symptoms. These results suggest that a tomato-rich diet may have a beneficial effect on the prevention of depressive symptoms. Further studies are needed to confirm these findings.